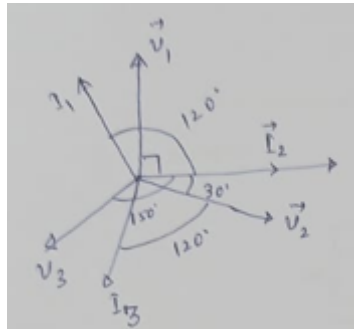


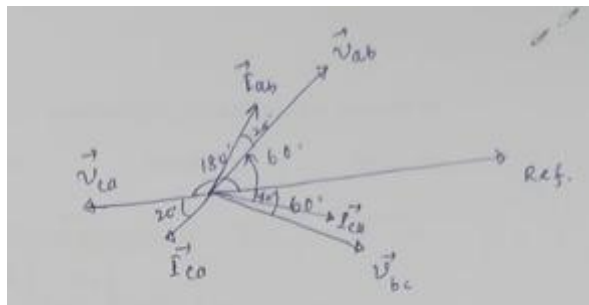
Answer Key Tutorial 7

1. Phase currents: $I_{AB} = 22\angle 45^\circ A$, $I_{BC} = 22\angle -75^\circ A$, $I_{CA} = 22\angle -195^\circ A$
 Line currents: $I_a = 38.105\angle 15^\circ A$, $I_b = 38.105\angle -105^\circ A$, $I_c = 38.105\angle -225^\circ A$

2. (a) $I_1 = 6\angle 120^\circ A$, $I_2 = 6 A$, $I_3 = 6\angle -120^\circ A$
 (b) $I_{yx} = 0 A$
 (c) $V_{cd} = 415.69\angle 60^\circ V$
 (d)

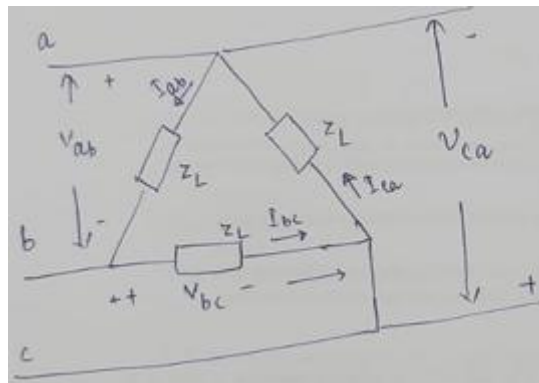


3. (a) $V_{ab} = 240\angle 60^\circ V$, $V_{bc} = 240\angle -60^\circ V$, $V_{ca} = 240\angle -180^\circ V$
 $I_{ab} = 6\angle 80^\circ A$, $I_{bc} = 6\angle -40^\circ A$, $I_{ca} = 6\angle 200^\circ A$



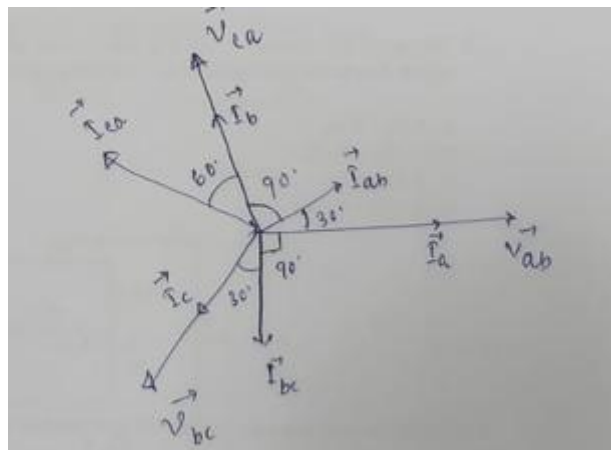
- (b) $Z_{ab} = 37.59 - j13.68 \Omega$
 (c) $I_a = 10.39\angle 50^\circ A$, $I_b = 10.39\angle -70^\circ A$, $I_c = 10.39\angle -190^\circ A$

4. (a)



(b) Phase currents: $I_{AB} = 12\angle 30^\circ A$, $I_{BC} = 12\angle -90^\circ A$, $I_{CA} = 12\angle 150^\circ A$
 Line currents: $I_a = 20.78 A$, $I_b = 20.78\angle -120^\circ A$, $I_c = 20.78\angle 120^\circ A$

(c)



(d) 7482.46 Watt

5. Phase currents: $I_{AB} = 5.47\angle -18.43^\circ A$, $I_{BC} = 5.47\angle -138.43^\circ A$,
 $I_{CA} = 5.47\angle 101.57^\circ A$

Line currents: $I_a = 9.48\angle -48.43^\circ A$, $I_b = 9.48\angle -168.43^\circ A$, $I_c = 9.48\angle 71.57^\circ A$

6. $V_{Load,line} = 158.99\angle 1.68^\circ V$

7. $S = (3946.67 + j5261.78) VA$